

P_C4H34

SAP Certified Professional - Developer - SAP Commerce Cloud

SAP CERTIFICATION TASKS

Course Version: 2601

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	<i>Example text</i>
Window title	<i>Example text</i>

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Instructions

In this system-based assessment, you will be asked to complete several tasks in an SAP system.

Data Entry Accuracy

Some tasks will require you to enter your group number and other data values. **It is critical that these values are entered exactly as specified** or you risk losing points. Pay close attention to:

- Correct capitalization of letters
- Empty spaces and all other symbols must be maintained exactly as stated.
- When entering your group number, replace redundant hashtags with leading zeros.

Example:

Your group number is 2

Input Value	Correct	Incorrect
Admin_##	Admin_02	admin_02, Admin 02
New Lead-####	New Lead-0002	New Lead-2, New Lead0002

Please make a note of your group number when you enter the assessment system shortly.

SAP Joule for Consultants

During a system-based assessment, you can use open-book resources, including SAP Joule for Consultants. To access it:

- Select SAP Joule for Consultants from the systems menu, along with other system options, before starting the assessment.
- It opens in a separate browser window at no cost.
- SAP Joule for Consultants remains available for the entire duration of your assessment.

For all other questions, please refer to [FAQs](#).

When you have finished reading these task instructions and are ready to begin the timed assessment, please return to the browser where you downloaded this document and follow the instructions there.

Good luck!

Resources and System Access

You'll take on the role of various employees of a fictional company using SAP Commerce Cloud in this exam. Each time you're asked to take on a role, you'll be given different tasks to test your understanding of SAP Commerce Cloud.

In this section, you'll be given information about resources you may use during the exam and how to set up the SAP Commerce Cloud instance you'll be using. So, read this section carefully.

Please note:

- You are allowed and invited to use different knowledge resources, such as the SAP Help Portal (help.sap.com), SAP Learning Site (learning.sap.com), or any other.
- **In particular**, you are encouraged to use [SAP Joule for Consultants](#) or the [Standard SAP Joule Client](#) (if you have access) to complete any tasks during the exam.
- Stick exactly to the defined names and parameters in the tasks. The validation of tasks is based on these requirements, and if they are not met, tasks cannot be validated as passed successfully.
- If parameters are not explicitly specified in the task or identifiable from the context, these can be chosen freely (e.g., the name of a cost center to be created).
- **Before you exit the exam, verify that the SAP Commerce Server is running. If it isn't, start the server and wait for it to finish starting up.**

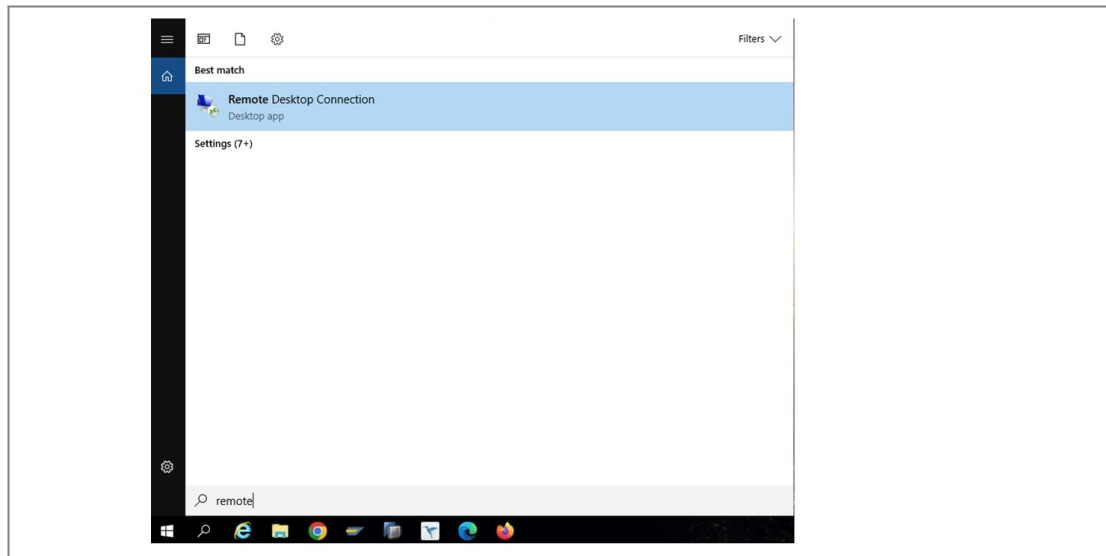
System Setup Guide (SSG) for the Certification

It is required to connect to the final Linux VM server and provide more credentials as in the following table:

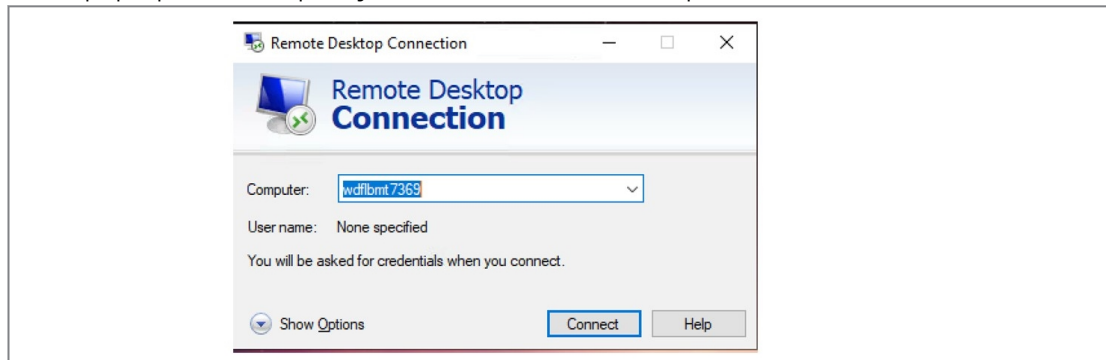
Linux VM	User Name	Password
wdf1bmt7369	c4hdevcert	c4h34com123

The concrete steps are described here:

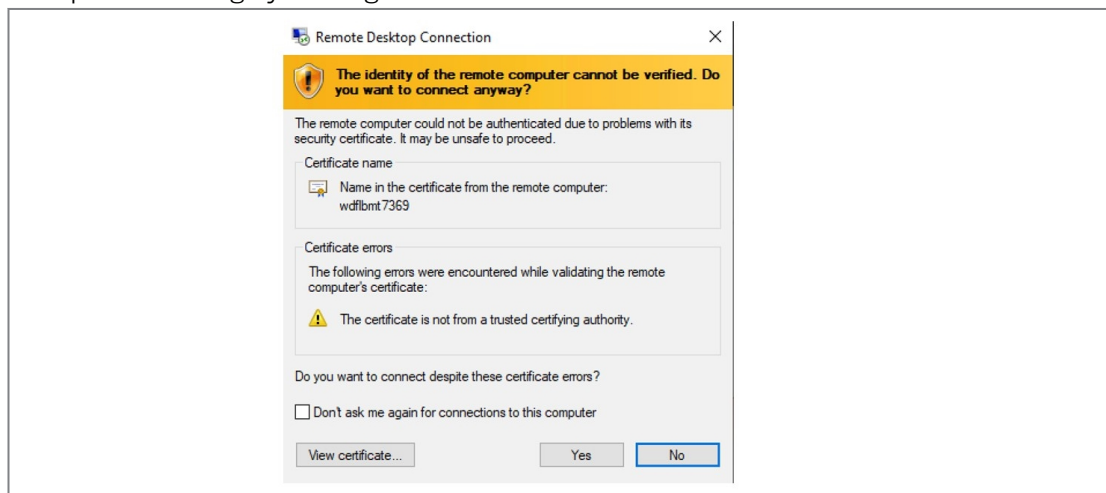
1. Once you're on the SAP Windows VM, click on the search icon besides the windows icon bottom left, and search for "remote" as follow and hit the "Remote Desktop Connection" app:



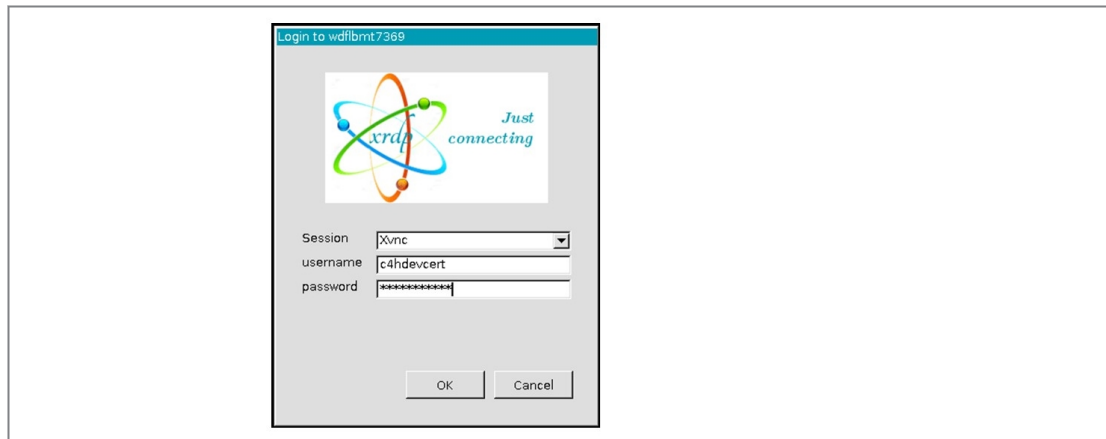
2. In the pop-up window, specify **wdflbmt7369** as the Computer name, then hit "Connect":



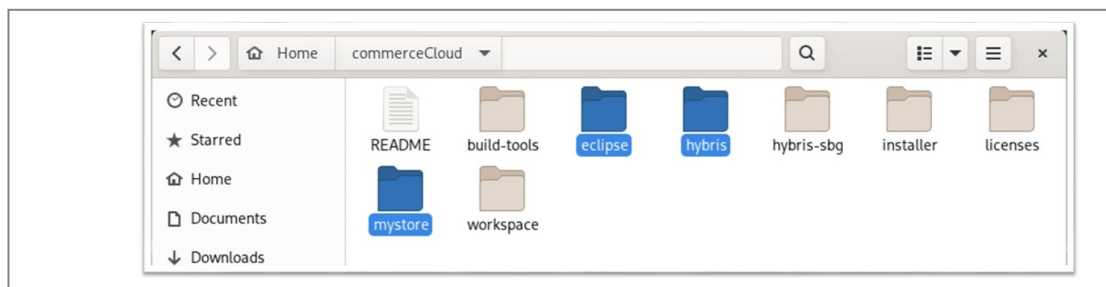
3. Accept the warning by clicking on "Yes":



4. In the next pop-up window, make sure you select "Xvnc" as the session type, enter the username "**c4hdevcert**" and password "**c4h34com123**", then hit "OK".

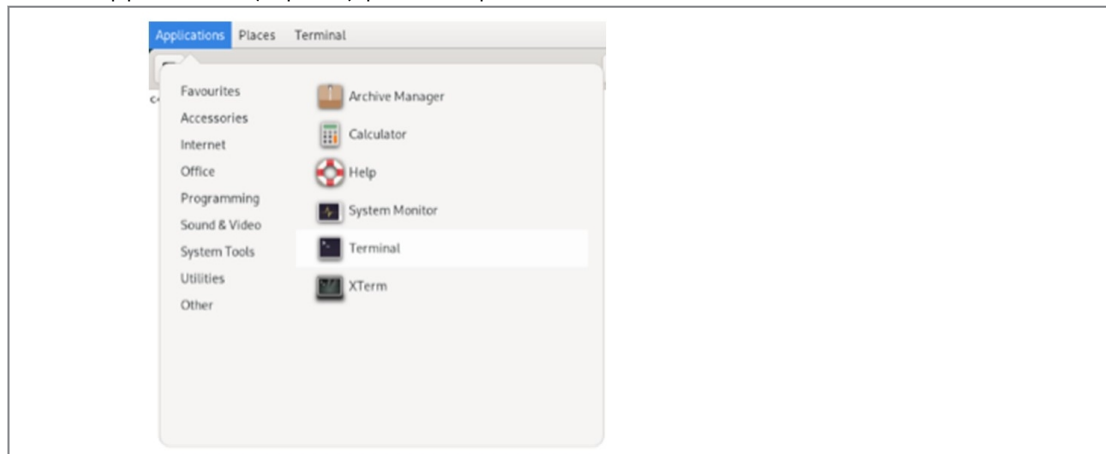


5. You're now on the Linux VM. Open the file manager (Applications > Accessories > Files). The resources for the certification are in the commerceCloud folder:
 - eclipse - the Eclipse IDE. (Visual Studio Code is also available; see the instructions for starting it later)
 - hybris – the commerce server installation.
 - mystore – the composable storefront installation.

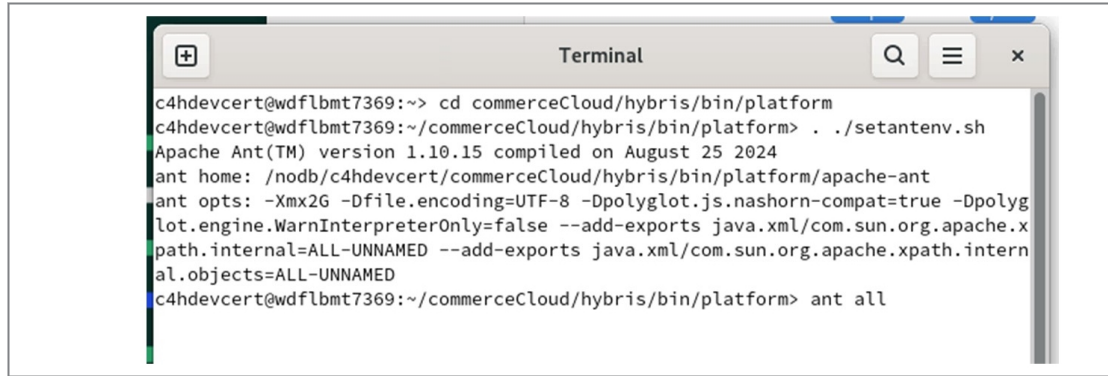


Using the command line tool:

1. Select Applications (top left) | Utilities | Terminal to start the Terminal console.



2. Go to the folder "commerceCloud/hybris/bin/platform/", there you can execute `./setantenv.sh` and `ant all` (You don't need to do this right away).

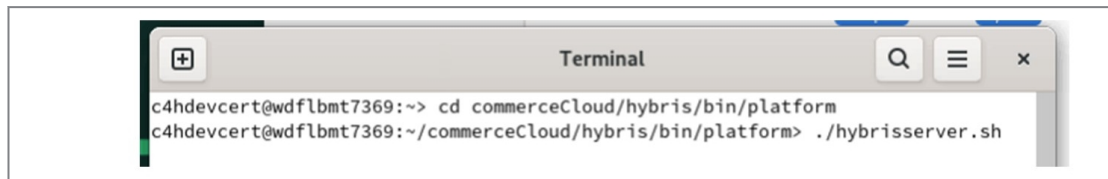


```

c4hdevcert@wdflbmt7369:~> cd commerceCloud/hybris/bin/platform
c4hdevcert@wdflbmt7369:~/commerceCloud/hybris/bin/platform> ./setantenv.sh
Apache Ant(TM) version 1.10.15 compiled on August 25 2024
ant home: /nodb/c4hdevcert/commerceCloud/hybris/bin/platform/apache-ant
ant opts: -Xmx2G -Dfile.encoding=UTF-8 -Dpolyglot.js.nashorn-compat=true -Dpolyglot.engine.WarnInterpreterOnly=false --add-exports java.xml/com.sun.org.apache.xpath.internal=ALL-UNNAMED --add-exports java.xml/com.sun.org.apache.xpath.internal.objects=ALL-UNNAMED
c4hdevcert@wdflbmt7369:~/commerceCloud/hybris/bin/platform> ant all

```

Alternatively, you can run **./hybrisserver.sh** (not now - it's only if you want to start the SAP Commerce Cloud Server).



```

c4hdevcert@wdflbmt7369:~> cd commerceCloud/hybris/bin/platform
c4hdevcert@wdflbmt7369:~/commerceCloud/hybris/bin/platform> ./hybrisserver.sh

```

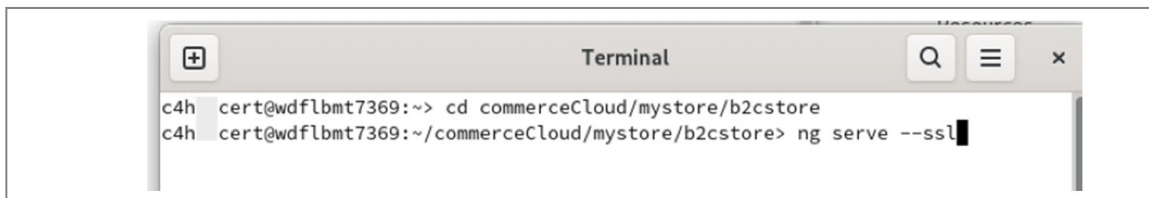


Note:

The Commerce server takes about 2-4 mins to start.

Starting Composable Storefront

For exercises that require the composable storefront, open a new Terminal window and navigate to the folder for the B2C installation using the commands **cd commerceCloud/mystore/b2cstore** and **ng serve --ssl** to start the storefront server.



```

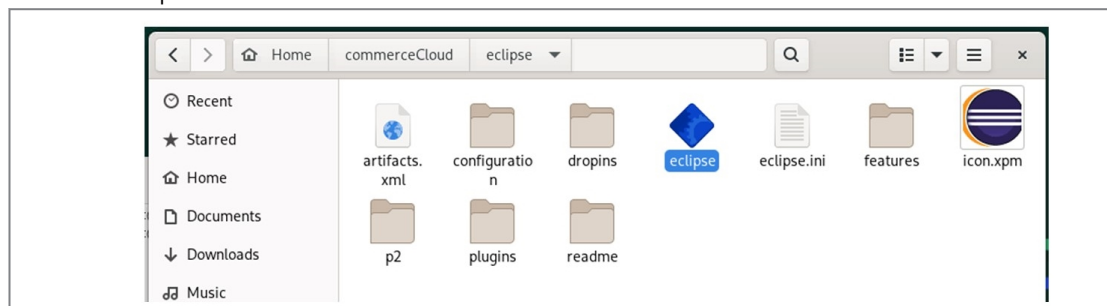
c4h cert@wdflbmt7369:~> cd commerceCloud/mystore/b2cstore
c4h cert@wdflbmt7369:~/commerceCloud/mystore/b2cstore> ng serve --ssl

```

Launching Eclipse IDE

When you need to use the Eclipse IDE:

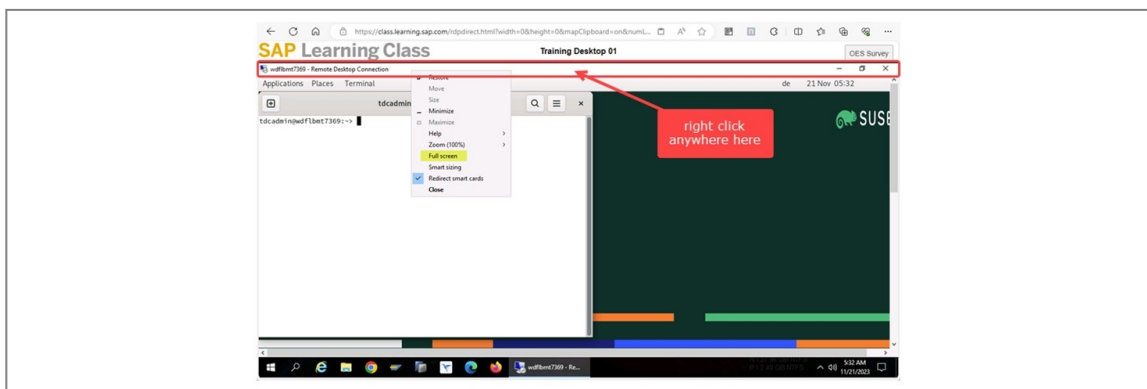
1. Open the file manager and navigate to the commerceCloud/eclipse folder, and double-click the eclipse file.



2. When Eclipse launches, the workspace opens with the commerce extensions projects already imported.



You can solve it by right-clicking on the xRDP window and choose either the "Full screen" mode or "Smart sizing", as shown in the screenshot below. If right-clicking isn't effective, first double-click on the xRDP window to bring up the window form of the remote desktop. Subsequently, right-click and select the aforementioned options.



Product Data Modeling

You are tasked with creating a toy store on SAP Commerce Cloud. Start by extending the default type system to support toy-related product types and attributes.

Prerequisites

Please start a terminal window and an IDE (either Eclipse or Visual Studio Code for simple editing), and proceed to complete the tasks.

Subtask 1: Create a New Extension

Requirements

Create a new extension **"datamodeling"** based on the **"yempty"** template with base package **"my.commerce.datamodeling"**. To support subsequent tasks, ensure the extension is included in the SAP Commerce Cloud build and startup. Build and import it into the Eclipse IDE (or Visual Studio Code).

Expected outcome

- A new extension is created with the specified name and base package.
- The extension is included in the SAP Commerce Cloud build and startup process.

Subtask 2: Define a New Type

Requirements

In the extension **datamodeling**, define a new type **Toy** that extends **Product**. Add an attribute named **"material"** of type `java.lang.String` to **Toy**, and ensure its value is persisted in the database.

Expected outcome

- **Toy** is defined in **datamodeling** with **Product** as its superclass.
- **Toy** includes the **"material"** attribute with persistent mapping.

Subtask 3: Create a Many-to-Many Relation

Requirements

In the **datamodeling** extension, create a non-localized many-to-many relation between **Toy** and **User**. Use **Toy2DesignerRelation** as the relation code, and apply **toys** as the qualifier for the **Toy** side and **designers** as the qualifier for the **User** side.

Expected outcome

- The many-to-many relationship is defined with the specified qualifiers.

Subtask 4: Build and Update SAP Commerce System

Requirements

Build, start, and update the SAP Commerce system so that the Toy type and the Toy-to-Designer relation defined in the previous tasks are available at runtime.

Expected outcome

- The SAP Commerce system is built and started successfully.
- The Toy type and the Toy-to-Designer relation are available (for example, in HAC or Backoffice).

Subtask 5: Create and Configure a New Toy**Requirements**

It's time to test the new type. Log in to Backoffice at [https://localhost:9002/backoffice/](https://localhost:9002/backoffice/admin/nimda/) (admin/nimda); in electronicsProductCatalog (Staged) version, create a new toy with:

- code: SAP001
- material: Wood

Additionally, to enable classification features for the toy products, create a classifying category in "ElectronicsClassification:1.0" with the identifier "safety" and a String classification feature "safetywarningcode" to render safety warnings, attach this classification feature to the SAP001 toy, and set safetywarningcode to "Toy01".

Expected outcome

- A new toy exists in electronicsProductCatalog (Staged) with code SAP001 and material Wood.
- Classification "safety" exists under "ElectronicsClassification:1.0" with the classification feature: "safetywarningcode".
- The SAP001 toy has the safetywarningcode value "Toy01".

Services

Nowadays, tax adjustments are often required due to policy changes - for example, new regional tax rates or updates to whether shipping is taxable. Your task is to develop services that identify the relevant taxes and update them accordingly.

Prerequisites

Please start an IDE (either Eclipse or Visual Studio Code for simple editing). You should see that an extension named base (under custom) has already been imported. This extension includes the interfaces and base classes required for the upcoming tasks.

Next, open a terminal window and proceed with the tasks.

Subtask 1: Develop a DAO Class

Requirements

In the base extension, CustomTaxDao interface defines a method to fetch all TaxModel instances whose value is greater than or equal to the given rate.

List <TaxModel> findTaxesByLeastRate(double rate);

You must implement a DAO class (named DefaultCustomTaxDao) that implements the CustomTaxDao interface. In the findTaxesByLeastRate method, use the FlexibleSearchService to retrieve all TaxModel instances whose **value** is greater than or equal to the provided rate. Register the DAO as a Spring bean with the id "customTaxDao".

As a good practice, do not add this DAO to the base extension; instead, create a new extension and implement the DAO there.

Expected outcome

- Create a new extension that includes a DAO class implementing the CustomTaxDao interface.
- The DAO uses the FlexibleSearchService to retrieve all related TaxModel instances and is registered as a Spring bean with the id "customTaxDao".

Subtask 2: Develop a Tax Service Class

Requirements

In the base extension, CustomTaxService interface defines a method as follows:

public int updateTaxRate(Double leastRate, int increasedBy);

This method updates all tax items whose value is at least leastRate (the first parameter) by adding increasedBy (the second parameter) to each value, and returns the number of tax items updated.

You need to create a DefaultCustomTaxService class that implements the CustomTaxService interface in the previously created extension. In the concrete implementation of the updateTaxRate() method, first call the DAO from the previous task to retrieve tax entries, update their values, and persist the changes using ModelService.

Finally, register the `DefaultCustomTaxService` as a Spring bean with the id `"customTaxService"`.

Expected outcome

- Develop a Tax Service named `DefaultCustomTaxService`, an implementation of `CustomTaxService`, which uses the DAO created in the previous task to retrieve relevant tax items, update them by adding the specified value, and persist changes via `ModelService`.
- The service is registered as a Spring bean with the id `"customTaxService"`.

Subtask 3: Develop an Interceptor

Requirements

For tax updates, customers require that the tax value must not exceed 100. Enforce this by implementing a `PrepareInterceptor` (in the previously created extension) that runs before persistence via `ModelService`. The interceptor should inspect updated `TaxModel` instances and change any value greater than 100 to 100. Register the interceptor as a Spring bean with the id `"taxPrepareInterceptor"` and map it (with the mapping id `"taxPrepareInterceptorMapping"`) accordingly.

Expected outcome

- Implement a `PrepareInterceptor` for the `TaxModel` that executes before `ModelService` persists it. If the tax value is greater than 100, set it to 100.
- Register the interceptor as a Spring bean with id `"taxPrepareInterceptor"` and configure it to apply only to the `Tax` type via the interceptor mapping with the specified id.

Order Management

Customers expect SAP Commerce Cloud to integrate with a backend ERP. On successful order placement, synchronize the order with the ERP for downstream processing. As the developer, implement the necessary order process changes to support this integration.

Prerequisites

Ensure that the SAP Commerce Server is stopped. Open Eclipse (or Visual Studio Code for simple editing) and a terminal window to complete the tasks:

Subtask 1: Replace the Default Order Fulfillment Extension

Requirements

As a preparation, create a new extension based on the `yacceleratorfulfilmentprocess` template, choosing any extension name and base package. Ensure the server build includes the new extension and replaces the `yacceleratorfulfilmentprocess` extension, then import the new extension into Eclipse (or Visual Studio Code for simple editing). In the new extension, locate the predefined order fulfillment process definition xml file and change the fulfillment process template name to "custom-order-process."

Expected outcome

- A new extension based on `yacceleratorfulfilmentprocess` exists
- The order fulfillment process template is renamed to "custom-order-process".
- The new extension is included in the build in place of `yacceleratorfulfilmentprocess`.

Subtask 2: Customize the Order Fulfillment Process

Requirements

In the new extension, implement a new action class that extends `AbstractProceduralAction<OrderProcessModel>` and publishes an `OrderSyncToERPEvent` (`org.training.examservices.events` in the pre-defined base extension) for the current order. Expose this action as a Spring bean with the id `orderSyncToERPAction`.

Next, extend the custom-order-process by inserting a new action node named `orderSyncToERP` between the existing `sendOrderCompletedNotification` node and the success end node. Relate the `orderSyncToERP` node to the `orderSyncToERPAction` bean.

Expected outcome

- A new `AbstractProceduralAction`-based class that publishes `OrderSyncToERPEvent`.
- A Spring bean named `orderSyncToERPAction` that references the custom action class.
- The `orderSyncToERP` node between `sendOrderCompletedNotification` and success in the custom-order-process template, wired to the `orderSyncToERPAction` bean.

Subtask 3: Implement an Event Listener

Requirements

The remote ERP is exposed via the MockERPSystem interface (in the base extension) and configured in Spring as the bean "mockERPSystem", with a method:

```
public void processOrder(OrderModel order);
```

Implement an event listener for OrderSyncToERPEvent that, when triggered, extracts the current order from the event and calls mockERPSystem.processOrder(order). Register the listener as a Spring bean with the id "orderSyncToERPEventListener".

Expected outcome

- A Spring bean orderSyncToERPEventListener that handles OrderSyncToERPEvent by retrieving the Order from the event and invoking mockERPSystem.processOrder(order).

Subtask 4: Apply the Custom Order Process Template

Requirements

Start the server with your changes from the previous steps. Then in Backoffice (log in with admin|nimda), locate the Electronics Store (BaseStore). It is currently configured to use the default order fulfilment process template. Update its configuration to enable and use the custom order process instead of the default.

Expected outcome

- The Electronics Store is configured to use the custom order process template named "custom-order-process".

Unit 1

Task 4

Commerce

You are part of the Backoffice team for an e-commerce site served at: <https://localhost:4200/USD/en/electronics-spa/>, please complete some typical daily tasks for configuring search experience as well as marketing campaigns such as coupons and promotions

Prerequisites

- Ensure the SAP Commerce Cloud instance and the B2C composable storefront are running.
- Access Backoffice at <https://localhost:9002/backoffice/> and log in as admin (password: nimda) to complete the tasks.

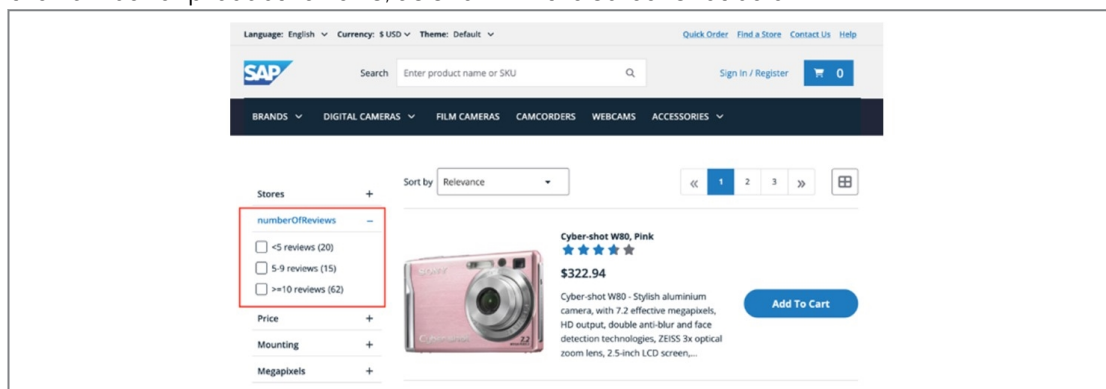
Subtask 1: Set up and Configure a Search Profile

Requirements

For testing purposes, create a new search profile directly on the electronics product catalog online version. Use the code “cxSearchProfile” and activate it with the online product catalog version for the site <https://localhost:4200/USD/en/electronics-spa/>.

Within this search profile, configure the following two requirements:

1. Add a new facet named “numberOfReviews” to the global category to filter products by the number of product reviews, as shown in the screenshot below:



Please make sure the following details:

- Attribute linkage: facet must be related to the Product type attribute numberOfReviews; note the attribute type int and create a corresponding indexed property (also named numberOfReviews) as preparation.
- Facet values: three values for ranges: <5 reviews, 5-9 reviews and >=10 reviews.
- Behavior: all three facet values must be combinable (for example, selecting “>=10 reviews” can be combined with the other two ranges).
- Placement: display the facet between the Stores and Price facets (do not change the configuration of the existing Stores and Price facets).

- For Digital Compacts at <https://localhost:4200/USD/en/electronics-spa/en/USD/c/576>, boost all Canon cameras (based on the indexed property "brandName") by increasing their weighting score by 30%.

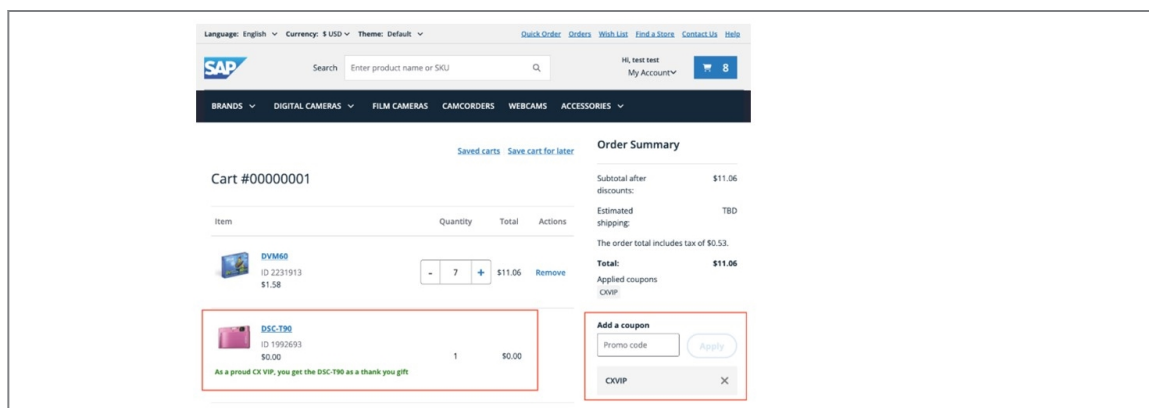
Expected outcome

- A new search profile with the code cxSearchProfile is created and activated for the e-commerce site <https://localhost:4200/USD/en/electronics-spa/>.
- The category pages display a new facet named numberOfReviews with three values: <5 reviews, 5-9 reviews, >=10 reviews, all of which are combinable.
- The facet is properly linked to Product.numberOfReviews via an appropriately prepared indexed property named numberOfReviews.
- The numberOfReviews facet is positioned between the Stores and Price facets without changing those existing default facets.
- On the Digital Compacts page (c/576), Canon cameras (brandName = "Canon") receive a 30% boost in their weighting/ranking score

Subtask 2: Create and Apply a Coupon and a Promotion

Requirements

Create a new promotion with the code cxCouponPromotion and publish it to the site at <https://localhost:4200/USD/en/electronics-spa/>;



ensure the promotion is part of the couponPromotionGroup and is the first promotion in the group to be evaluated, and that it should only activate for customers who belong to the CXVIPGroup user group; if an eligible customer's cart total is at least \$10, they can redeem the coupon CXVIP to receive a free gift: the DSC-T90 pink camera (product code: 1992693); the CXVIP coupon may be redeemed no more than twice per customer and no more than 250 redemptions in total.

Expected outcome

- A promotion with code cxCouponPromotion is created, configured, published and can be tested on the site at <https://localhost:4200/USD/en/electronics-spa/>.
- The promotion is associated with the couponPromotionRuleGroup and is the first promotion evaluated in the promotion evaluation order.
- The promotion is defined with 3 conditions (either all conditions in 1 group, or no group at all) and 1 action:

- Condition 1: Restricted to customers in the CXVIPGroup user group.
- Condition 2: Cart total is at least \$10.
- Condition 3: Coupon code CXVIP.
- Action: Award the free gift DSC-T90 pink camera (product code 1992693).
- Redemption limits are enforced: the CXVIP coupon can be redeemed at most twice per customer and the total redemptions across all customers cannot exceed 250.

Web Content Management

You are the technical support for managing the web content of the e-commerce site served at <https://localhost:4200/USD/en/electronics-spa/>, working with SmartEdit and Backoffice to perform daily tasks.

Prerequisites

Verify that the SAP Commerce Cloud server and the B2C Composable Storefront are running. Launch Firefox and navigate to the following URLs to complete the tasks (credentials provided):

1. SmartEdit: <https://localhost:9002/smartedit> (admin/nimda)
2. Backoffice: <https://localhost:9002/backoffice> (admin/nimda)



Note:

For testing purposes, use only the **Staged** version catalog in all upcoming tasks; no synchronization is required.

Subtask 1: Set up a CMS Component

Requirements

In the Spartacus Electronics Site homepage (Staged), locate the Section1Slot-Homepage slot and add a new cms component of type CMSTitleParagraphComponent to that slot with the following properties:

- name: welcomeTitleParagraph
- title: any non-null value
- content: any non-null value



Note:

CMSTitleParagraphComponent has just been implemented. Ensure that this cms component type is allowed in the Section1Slot-Homepage slot before adding it.

Expected outcome

- A CMSTitleParagraphComponent exists with the specified attributes and values.
- The CMSTitleParagraphComponent is added to the content slot "Section1Slot-Homepage" on the Spartacus Electronics Site homepage (Staged).

Subtask 2: Personalize a CMS Component

Requirements

Ensure that the newly added CMSTitleParagraphComponent is visible only to users in the eduUserGroup.



Note:
The eduUserGroup already exists in the system.

Expected outcome

- The newly added CMSTitleParagraphComponent is restricted to display only to users in the eduUserGroup.

Integration

You are part of the Integration team for an e-commerce site served at: <https://localhost:4200/USD/en/electronics-spa/>, please complete routine daily tasks to customize various integration options.

Prerequisites

- Ensure the SAP Commerce Cloud instance is running.
- Access Backoffice at <https://localhost:9002/backoffice/> and log in as admin (password: nimda) to complete the tasks.
- Optional: A Bruno API Client (Top left | Applications | Programming | Bruno) is available for testing your Integration Object if needed.

Subtask 1: Set up an Integration Object

Requirements

Create a first-draft Integration Object (IO) to exchange Customer data with an external CRM. The IO should model the Customer data type and be named "CXCustomerIO".

It must expose the attributes "contactEmail", "name", and "uid"; the endpoint should be secured with Basic Authentication to enable a primary test; the data returned by the endpoint should be limited to Customer instances and their subtypes; and the IO must include a virtual attribute named "orderCount" that counts the orders associated with the customer (using the existing attribute orders).



Note:

To implement the virtual attribute logic, you may need to use the SAP Commerce Cloud Administration Console (<https://localhost:9002/> with admin/nimda) to create a new script.

Expected outcome

- "CXCustomerIO" created and active, mapping Customer data
- Exposed attributes: contactEmail, name, uid
- Endpoint secured with Basic Authentication and reachable for tests
- Data scope limited to Customer instances/subtypes only
- Virtual attribute "orderCount" implemented as the count of related orders (0 if none)

Subtask 2: Set up a Webhook Configuration

Requirements

Please create and configure a webhook setup that publishes customer data via the CXCustomerIO integration object to an external site.

As preparation, first create a consumed destination, name it "cxCertDestination" and link it to a unique URL generated at <https://webhook.site> by visiting the site and obtaining the URL. Configure the consumed destination with the necessary properties so that it can be used by a webhook configuration.



Note:

The <https://webhook.site> does not require a security check, so no authentication is needed for this test.

After the consumed destination is in place, create a webhook configuration that references both the CXCustomerIO integration object and the prepared consumed destination. Ensure that webhook messages are published to the consumed destination only when a customer is created or updated.

Expected outcome

- A consumed destination named "cxCertDestination" linked to the unique webhook.site URL is created.
- A webhook configuration references the CXCustomerIO integration object and the prepared destination.
- Webhook publishes only if a Customer is created or updated.

Submitting your Assessment

- Select the Confirm completion button to submit your assessment. This button is in the browser tab where you initially accessed the assessment.
- Please wait for the validation process to complete, this can take up to 15 minutes.
- Results will be displayed on learning.sap.com under My Certifications.